Hall Ticket Number:

MR-22

Code: B425CS5S

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Supplementary Examinations, December 2025 / January 2026

Software Engineering

(Common to CSE, CSB and CSD)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 60

Part-A (10 x 1 = 10 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	What are software engineering layers?	1	1	1	1
1. b)	What is process assessment?	1	1	1	1
1. c)	What is interface specification?	1	1	2	2
1. d)	Define requirements validation.	1	2	2	2
1. e)	What is class diagram?	1	1	3	3
1. f)	What is software architecture? Define.	1	1	3	3
1. g)	What is software testing?	1	1	4	4
1. h)	What is meant by debugging?	1	1	4	4
1. i)	What is software review?	1	1	5	4
1. j)	Define Risk Refinement.	1	2	5	4

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Illustrate in brief about that waterfall model.	5	2	1	1
2. b)	Explain about evaluation of software engineering methodologies.	5	2	1	1
	OR				
2. c)	Explain about software development process models.	5	2	1	1
2. d)	Demonstrate the various software myths.	5	2	1	1
3. a)	Explain how a software requirements document is structure?	5	3	2	2
3. b)	What are feasibility studies? Explain its types.	5	2	2	2
	OR				
3. c)	What are the desirable characteristics of a good software requirement specification document? Discuss.	5	3	2	2
3. d)	Explain about requirements management phases of requirement engineering process.	5	2	2	2
4. a)	Briefly explain about the design model.	5	3	3	3
4. b)	Model the class diagram and explain various relationships that exist between classes.	5	3	3	3
	OR				
4. c)	Construct software architecture. Explain why it may be necessary to design the system architecture before the specifications.	5	3	3	3
4. d)	Distinguish between sequence and collaboration diagrams.	5	4	3	3
5. a)	Explain in detail about black box testing techniques.	5	4	4	4
5. b)	Explain clearly about metrics for software quality.	5	3	4	4
	OR				
5. c)	Write short note on system testing.	5	2	4	4
5. d)	Discuss about metrics for testing in detail.	5	3	4	4
6. a)	Explain significance formal technical review.	5	2	5	4
6. b)	Examine the methods for Risk Projection.	5	4	5	4

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	OR				
6. c)	What types of risk occur during software development? Discuss.	5	2	5	4
6. d)	Summarize the role of formal technique reviews in achieving good quality software.	5	2	5	4

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome